

Majorana zero Edge States with longer-range interactions in a quantum Ising chain

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Major Project Report

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1 Introduction

In this report we will Study in depth about various 1-d spin models like the most basic Transverse field ising model, the X-Y model in transverse field and a three spin extension of the TFIM and study their phases and various properties. We will look at this model from the perspective of its majorana zero modes. We will also study several topological phase transitions that occur in this model. In this study the plan is to initially introduce the ising model and XY model and its diagonalization techniques. Its important to study such models to build strong theoretical framework for understanding quantum phase transitions and classify them according their properties and phases.

1.1 Background and History

In Late 1930s, Physicist Ettore Majorana predicted the existence of a new particle called majorana fermions. these fermions are their own anti particles. Research to detect such particles is ongoing via a extremely rare neutrino-less double beta decay. In recent years the research on this topic has gained momentum and majorana bound states are predicted to exist in semiconductor nanowires, magnetic atom chains and topological insulators. Majorana fermions also make good candidates of topological quantum computation since they don't interact with the environment. Also in quantum algorithms realised via exchange of majorana fermions and qubits made by majorana fermions are expected to be topologically protected from noise and decoherence. Hence small disturbances cannot decohere the qubit making it more robust for use in quantum computers.

2 1-D Transverse field Ising model

The Hamiltonian for 1D transverse field ising model is given by,[\[1\]](#)

$$H = -J \sum_{i=1}^N \sigma_i^x \sigma_{i+1}^x - h \sum_{i=1}^N \sigma_i^z \quad (1)$$

where, $h = Jg$, The parameter g effectively scales the strength of the magnetic field relative to the strength of the spin-spin interaction.

- Here, J is the strength of the spin-spin interactions, h is the strength of the transverse magnetic field, σ_i^x and σ_i^z are the Pauli matrices acting on the i -th spin, and the sum is taken over pairs of nearest-neighbor spins.
- The first term in the Hamiltonian represents the interaction between spins in the x -direction, and the second term represents the transverse magnetic field applied in the z -direction. The model exhibits quantum phase transitions as a function of the parameters J and h .
- Using Jordan Wigner and Bogoliubov De Genne transformations this model can be exactly solved (as we will show in subsequent sections), we can write, and the results can be found here . [\[2\]](#) and [\[1\]](#)

$$H^\pm = \sum_q \epsilon \left(\frac{2\pi q}{N} \right) \left(\chi_q^\dagger \chi_q - \frac{1}{2} \right) \quad (2)$$

where,

$$\epsilon(\alpha) = \sqrt{J^2 + h^2 - 2Jh \cos(\alpha)} \quad (3)$$

here, χ_q 's are fermionic operators obeying anticommutation relations. The difference between ground state energy and first excited state energy (minimum excitation energy) is given by, (energy gap = Δ)

$$\Delta = 2|J - h|$$

for $h = Jg$,

$$\Delta = 2J|1 - g| \quad \forall J > 0$$

At $g = 1$, this gap vanishes. This point is the Quantum phase transition point. It represents a phase boundary between the ordered and disordered phase.

- **Ordered Phase $g \ll 1$ (Ferromagnetic Phase):** When the spin-spin interaction (J) dominates over the transverse magnetic field (h), the system tends to align spins parallel to each other, resulting in a ferromagnetic phase. In this phase, spins tend to point in the same direction, and there is a net magnetization along the z -direction.
- **Disordered Phase $g \gg 1$ (Paramagnetic Phase):** When the transverse magnetic field (h) dominates over the spin-spin interaction (J), the spins are disordered and fluctuate, leading to a paramagnetic phase. In this phase, there is no long-range order, and the spins are not aligned.
- **Quantum Phase Transition:** There is a critical point in the phase diagram where the transition between the ordered and disordered phases occurs.
- **Unlike classical phase transitions that are driven by thermal fluctuations, quantum phase transitions are driven by quantum fluctuations ideally at $T=0$ K**

3 1-D XY Model in transverse field

The model we introduced in Eq.(1) can be generalized to more dimensions or we can add more interactions between spins in different directions. We study transverse-field XY model which is commonly formulated with spins that have quantum spin- $\frac{1}{2}$. In this context, each individual spin is described by a two-dimensional Hilbert space, and the Pauli matrices (σ^x , σ^y , σ^z) are used to represent the spin operators. We can add a interaction in y direction and such a model is given by,

$$H = \frac{J}{2} \sum_{i=1}^N \left[\left(\frac{1+\gamma}{2} \right) \sigma_i^x \sigma_{i+1}^x + \left(\frac{1-\gamma}{2} \right) \sigma_i^y \sigma_{i+1}^y + g \sigma_i^z \right] \quad (4)$$

The term $\frac{J}{2}$ represents the coupling constant, indicating the strength of interactions between neighboring spins. The summation $\sum_{i=1}^N$ encompasses all sites in the system (from $i=1$ to N). The terms $\frac{1+\gamma}{2}$ and $\frac{1-\gamma}{2}$ modulate the strength of these interactions, controlled by the anisotropy parameter γ . The anisotropy parameter γ determines the relative strength of the interactions in different directions.

- When $\gamma = 1$, the system is isotropic, with equally strong interactions in the x - and y -directions. - When $0 < \gamma < 1$, the system exhibits anisotropy, favoring interactions in the x -direction over the y -direction. - When $\gamma > 1$, anisotropy favors interactions in the y -direction over the x -direction.

The parameter γ allows for the tuning of the anisotropic behavior, influencing the competition between different types of interactions in the model. For the x -interaction term, $\sigma_i^x \sigma_{i+1}^x$ represents the product of x -components of spins at sites i and $i+1$. For the y -interaction term, $\sigma_i^y \sigma_{i+1}^y$ represents the product of y -components of spins at sites i and $i+1$. The term $h\sigma_i^z$ represents the influence of an external magnetic field h in the z -direction.

Similar to the model in (1) this model can also be solved exactly using certain diagonalization techniques as will be shown in sections below. Where we will solve this model. Given as below, the results used below can be found here([2])

$$H^{\pm} = \sum_q \epsilon \left(\frac{2\pi q}{N} \right) \left(\chi_q^{\dagger} \chi_q - \frac{N}{2} \right) \quad (5)$$

where,

$$\epsilon(\alpha) = \sqrt{(h - \cos \alpha)^2 + \gamma^2 \sin^2(\alpha)} \quad (6)$$

where, here $\pm$ above H indicates the hamiltonian can be written separately for even no of particles and odd no of particles. The χ_q 's are fermionic operators obeying anticommutation relations.

The phase diagram of this Hamiltonian will depend on the values of the parameters J , γ , and h . The key features of the phase diagram will include the nature of the ground state and any quantum phase transitions that may occur.

4 Jordan Wigner Transformation and Bogoliubov-de-gennes Transformation

What is J-W transformation?

- Solving 1D quantum spin models, the mapping of spin-1/2 Pauli operators to free fermionic operators is often done to simplify the mathematical description of the system.
- The Jordan-Wigner transformation allows one to represent spin operators in terms of fermionic creation and annihilation operators. These fermions are associated with each site of the original spin chain.[2]

$$\begin{aligned}
\sigma_j^+ &= \exp \left(i\pi \sum_{k=1}^{j-1} c_k^\dagger c_k \right) c_j^\dagger, \\
\sigma_j^- &= \exp \left(-i\pi \sum_{k=1}^{j-1} c_k^\dagger c_k \right) c_j, \\
\sigma_j^z &= 1 - 2c_j^\dagger c_j,
\end{aligned}
\quad \text{where } c_j^\dagger \text{ and } c_j \text{ are fermionic creation and annihilation operators.}
\tag{7}$$

What is BdG transformation? Once we obtain the hamiltonian in fermionic representation and then apply fourier transform to tranform it in terms of k and apply this BdG tranformation, we have then a diagonalized hamiltonian.

We can diagonalize each of the matrices (and thus the whole Hamiltonian) by means of a Bogoliubov transformation, which, in our notation using a new set of operators that are sometimes called bogoliubov quasiparticles which similar to the fermionic operators follow the anticommutation relations. The transformation is given by following equations,

$$\begin{bmatrix} c_k \\ c_{-k}^\dagger \end{bmatrix} = \begin{bmatrix} \cos(\phi_k) & \sin(\phi_k) \\ -\sin(\phi_k) & \cos(\phi_k) \end{bmatrix} \begin{bmatrix} \chi_k \\ \chi_{-k}^\dagger \end{bmatrix}
\tag{8}$$

5 Introduction to Majorana Fermions and Majorana Zero Modes

Majorana fermions are particles that are their own antiparticels.

Majorana fermions are quasiparticles that can form in superconducting systems within condensed matter systems. Proposed by Alexei Kitaev, the Kitaev chain is one theoretical model used to analyze Majorana fermions. Majorana zero modes (MZMs) are bound states with zero energy. Because of their non-Abelian statistics, MZMs may find use in quantum computing. This model sheds light on their behavior.

5.1 Fermions

Fermions are fundamental particles characterized by their half-integer spin. In the context of quantum mechanics, they are described by creation and annihilation operators denoted as $\hat{a}^\dagger$ and $\hat{a}$ respectively. These operators satisfy the following commutation relations:

$$\{\hat{a}^\dagger, \hat{a}\} = \mathbf{I}, \quad \{\hat{a}, \hat{a}^\dagger\} = 0, \quad \hat{a}^2 = 0 = (\hat{a}^\dagger)^2$$

Here, $\mathbf{I}$ is the identity operator. Fermions can occupy two states, conventionally denoted as $|0\rangle$ and $|1\rangle$, corresponding to zero and one occupation numbers respectively. The occupation number operator is defined as $\hat{n} = \hat{a}^\dagger \hat{a}$, and its action on the states is given by:

$$\hat{a}|0\rangle = 0, \quad \hat{a}^\dagger|1\rangle = 0, \quad \hat{a}|1\rangle = |0\rangle, \quad \hat{a}^\dagger|0\rangle = |1\rangle$$

5.2 Majorana Fermions

Majorana fermions are particles that are their own antiparticles. They follow Fermi-Dirac statistics and possess the property of being self-conjugate, meaning they are equal to their own Hermitian conjugate: $\hat{\gamma} = \hat{\gamma}^\dagger$.

the fermionic creation operator $\hat{a}^\dagger$ in terms of Majorana operators $\hat{\gamma}_1$ and $\hat{\gamma}_2$ as follows:

$$\hat{a}^\dagger = \frac{1}{2}(\hat{\gamma}_1 + i\hat{\gamma}_2)$$

some of the properties of Majorana fermions are:

$$\hat{\gamma}_1 = \hat{\gamma}_1^\dagger, \quad \hat{\gamma}_1^2 = \mathbf{I}, \quad \text{and} \quad \{\hat{\gamma}_1, \hat{\gamma}_2\} = 2\delta_{ij}$$

5.3 Majorana Zero Modes in the Kitaev Chain

The kitaev chain is given by,

$$H = \sum_{i=1}^N \left(-\mu \left(a_i^\dagger a_i - \frac{1}{2} \right) - w(a_i^\dagger a_{i+1} + a_{i+1}^\dagger a_i) + \Delta a_i a_{i+1} + \Delta^* a_{i+1}^\dagger a_i^\dagger \right) \quad (9)$$

The first parameter μ simply corresponds to the occupation energy of each site, which can host either 0 or 1 electron. The second parameter w is the hopping energy, with terms $a_i^\dagger a_{i+1}$ and $a_{i+1}^\dagger a_i$ corresponding to particle motion towards the left and right ends of the chain. Finally, the third parameter, $\Delta = \Delta \exp i\theta$, corresponds to the induced superconducting pairing. This is the energy it takes to break up the Cooper pairs, created by pairwise operators $a_i^\dagger a_{i+1}^\dagger$ and annihilated by $a_i a_{i+1}$, with θ being the spontaneously chosen coherent phase of the superconductor.

After writing this in majorana basis and diagonalizing the hamiltonian we get following form, the results used below can be found here [3]

$$H = 2w \sum_{i=1}^{N-1} \left(\tilde{a}_i^\dagger \tilde{a}_i - \frac{1}{2} \right) \quad (10)$$

where, $\tilde{a}_i = (\gamma_{2i} - i\gamma_{2i+1})/2$

In this instance, we initiated with N fermionic operators in the Hamiltonian but are now left with $N - 1$. This stems from the pairing of Majorana modes with odd parity, thus excluding the two modes at the end of chain from the Hamiltonian. Consequently, $[H, \gamma_1] = [H, \gamma_{2N}] = 0$, rendering them energetically neutral. This gives rise to what is termed as Majorana zero modes.

A new fermionic operator $\tilde{a}_0 = (\gamma_1 + i\gamma_{2N})/2$ is defined, composed of MZMs at the chain's two ends. This facilitates the definition of degenerate computational basis states $\tilde{a}_0|0\rangle = 0$ and $\tilde{a}_0^\dagger|0\rangle = |1\rangle$, enabling the storage and manipulation of quantum information. As intended, the qubit has been effectively divided into two spatially segregated modes.

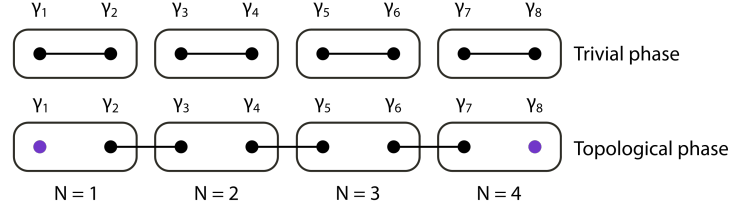


Figure 1: Two ways to pair the the Majorana modes in a Kitaev chain. at each site we have the two Majorana modes are γ_{2N} and γ_{2N+1} . In the trivial phase, all the Majorana pairs are paired, while in the topological phase, the pairing is offset by one, leaving unpaired modes γ_1 and γ_8 at the two end.

6 Majorana zero Edge States with longer-range interactions in a quantum Ising chain

Kitaev's model is the fermionized version of the familiar one dimensional transverse field Ising model (TFIM) which is one of the simplest models of quantum criticality.

However, in a quantum wire, the range of Cooper pairing and the electron wave function hybridization will be finite, and the impact of these longer range interactions needs to be taken into consideration. This paper [4] aims to investigate the impact of these extended interactions. When we perform this in the framework of another precisely solvable model, we discover that the longer distant interactions produce a rich phase diagram.

6.1 The Hamiltonian and the phase diagram

The three spin extension of TFIM is given by hamiltonian,

$$H = - \sum_i (g\sigma_i^x + \lambda_2\sigma_i^x\sigma_{i-1}^z\sigma_{i+1}^z + \lambda_1\sigma_i^z\sigma_{i-1}^x) \quad (11)$$

The σ 's are the standard pauli matrices.

Diagonalizing this hamiltonian requires multiple steps, first is using the Jordan-Wigner transformation. See appendix (10) for complete solution of this.

$$\begin{aligned} \sigma_i^x &= 1 - 2c_i^\dagger c_i, \\ \sigma_i^z &= \prod_{j<i} (1 - 2c_j^\dagger c_j) (c_i + c_i^\dagger), \end{aligned} \quad (12)$$

the fermionized form after J-W looks something like this,

$$H = -g \sum_{i=1}^N (1 - 2c_i^\dagger c_i) - \lambda_1 \sum_{i=1}^{N-1} (c_i^\dagger c_{i+1} + c_i^\dagger c_{i+1}^\dagger + h.c) - \lambda_2 \sum_{i=2}^{N-1} (c_{i-1}^\dagger c_{i+1} + c_{i+1} c_{i-1} + h.c) \quad (13)$$

after imposing periodic boundary condition, the hamiltonian can be diagonalized using bogoliubov-de-gennes transformation giving us,

$$H = \sum_k \epsilon_k \left(\eta_k^\dagger \eta_k - \frac{1}{2} \right) \quad (14)$$

where η_k 's are suitable linear combination of the original J-W fermions in the momentum space,

the spectra of excitations is given as,

$$\epsilon_k = \pm \sqrt{1 + \lambda_1^2 + \lambda_2^2 + 2\lambda_1(1 - \lambda_2) \cos k - 2\lambda_2 \cos(2k)} \quad (15)$$

- We know for ising model in transverse field the gap closes at the brillouin zone boundaries i.e at $k = \pm\pi$ at the self dual points $\lambda_1 = 1$ and $\lambda_2 = 0$
- For this model the gap can collapse at $k = 0$ and $k = \arccos(\lambda_2/2)$ for $\lambda_2 = -1$ and $0 < \lambda_1 < 2$
- for the point b in the figure $\lambda_2 = -1$ and $\lambda_1 = 0$ the gap collapses for $k = \pm\pi/2$
- the point 'a' is a special multicritical point where $\lambda_1 = 0$ and $\lambda_2 = 1$
- The Fig. (2) shows that there are three zones of $n = 0$, $n = 1$, and $n = 2$ zero modes. The lines dividing these regions are quantum critical lines, with the exception of the topological transition point 'a'.
- In the original spin representation, the region $\lambda_2 > 1 + \lambda_1$ is an ordered phase, and the gap at $k = 0$ collapses at the boundary of this region, which is marked by a critical line.
- The disordered phase area $\lambda_2 < 1 - \lambda_1$ has a critical line as its boundary, where the gap closes at $k = \pi$.
-

6.2 Zero Energy Edge States

So What are MZMs?

- A Majorana zero edge state is a fermionic operator γ that squares to 1 and, as a result, commutes with the system's Hamiltonian H because of its zero energy: Since $\gamma^2 = 1$ and $[\gamma, H] = 0$, γ is fermionic.
- A Majorana fermion operator is any operator that meets the first two criteria. And It is a Majorana zero mode operator, if it also fulfils the third requirement. Such operators imply the existence of a degenerate space of ground states, which is capable of storing quantum information.
- From the diagonalization of the BdG hamiltonian we can see majorana zero modes only for open boundary condition.

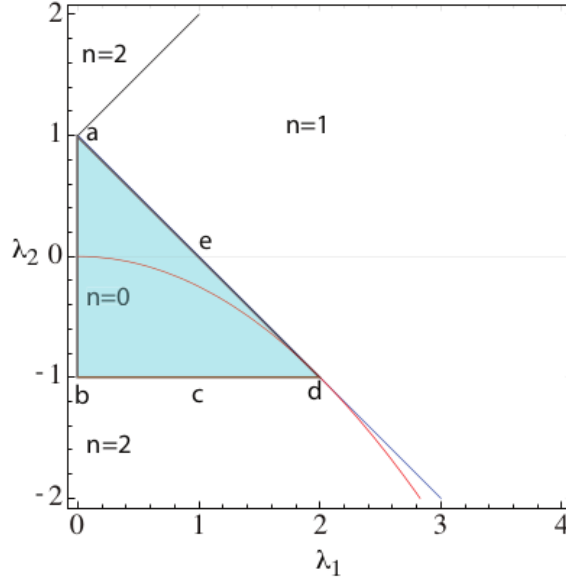


Figure 2: Phase diagram of the model proposed in Eq.(??) where we can see three different regions separated by critical lines.

- Assuming open boundary conditions the equations are,

$$\begin{pmatrix} \tilde{h} & \Delta \\ -\tilde{\Delta} & -\tilde{h} \end{pmatrix} \begin{pmatrix} \vec{u}_n \\ \vec{v}_n \end{pmatrix} = E_n \begin{pmatrix} \vec{u}_n \\ \vec{v}_n \end{pmatrix} \quad (16)$$

where the submatrices are,

$$\begin{aligned} \tilde{h}_{ij} &= \lambda_1 (\delta_{j,i+1} + \delta_{j,i-1}) + \lambda_2 (\delta_{j,i+2} + \delta_{j,i-2}) - 2\delta_{ij}, \\ \tilde{\Delta}_{ij} &= -\lambda_1 (\delta_{j,i+1} - \delta_{j,i-1}) - \lambda_2 (\delta_{j,i+2} - \delta_{j,i-2}) \end{aligned} \quad (17)$$

- Using the above equations we can numerically simulate and find the edge states for various values of λ_1 's and λ_2 's
- One of the plots for $n=2$ region where $\lambda_2 = -1.2$ and $\lambda_1 = 1$ as shown in Fig.(3).
- Another one of the plots given in Fig.(4) shows the majorana modes for the region where $\lambda_2 = 1.5$ and $\lambda_1 = 0.05$ in which we see two modes at the end of the chain.
- Another region where we can see one majorana mode at the end of the chain is $n = 1$ region in fig(2) can be plotted. as shown in 5

7 Unbroken Unitary time reversal symmetry

Upon applying fourier transformation to Eq.(13) we get,

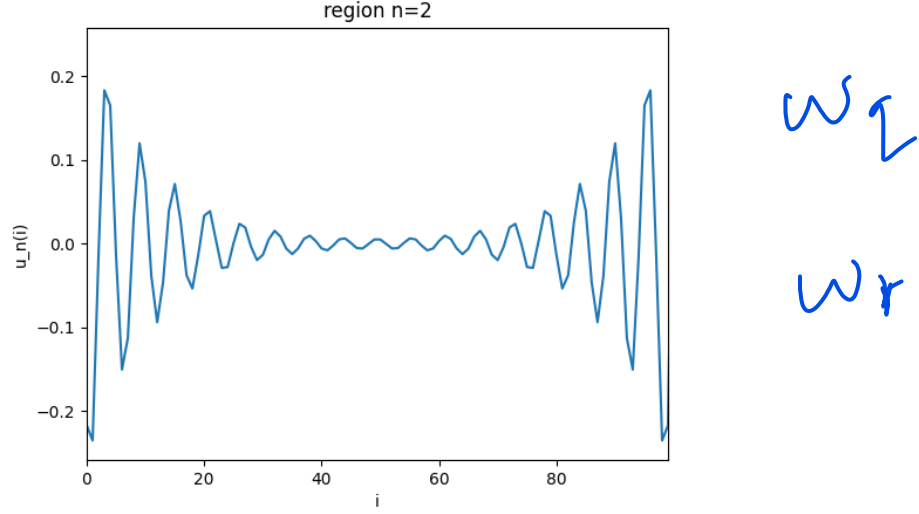


Figure 3: the two majorana zero modes for $\lambda_2 = -1.2$ and $\lambda_1 = 1$

$$H = \sum_k (2 - 2\lambda_1 \cos k - 2\lambda_2 \cos 2k) c_k^\dagger c_k + \sum_k \left(i\lambda_1 \sin k c_k^\dagger c_{-k}^\dagger + i\lambda_2 \sin 2k c_k^\dagger c_{-k}^\dagger + \text{h.c.} \right) \quad (18)$$

The BdG Hamiltonian, with the following form, controls the BCS quasiparticles' behavior at each momentum k :

$$H_{BdG}(k) = \begin{pmatrix} \epsilon_k - \mu & i\Delta(k) \\ -i\Delta(k) & \mu - \epsilon_k \end{pmatrix} \quad (19)$$

where,

$$\begin{aligned} \epsilon_k &= -2\lambda_1 \cos k - 2\lambda_2 \cos 2k, \\ \Delta(k) &= \lambda_1 \sin k + \lambda_2 \sin 2k \end{aligned} \quad (20)$$

What is time reversal symmetry?

- When the hamiltonian is real we can say that it follows time reversal symmetry.
- Put another way, the underlying physics that a Hamiltonian describing a physical system with TRS describes would not change even if you were to reverse the direction of time in that system. This suggests that time reversal preserves the symmetry of the principles of physics.
- Mathematically if $\hat{H}$ represents the hamiltonian operator and $\hat{T}$ represents time reversal operator then,

$$\hat{T}\hat{H}\hat{T}^{-1} = \hat{H} \quad (21)$$

means applying the TRS operator leaves the hamiltonian unchanged

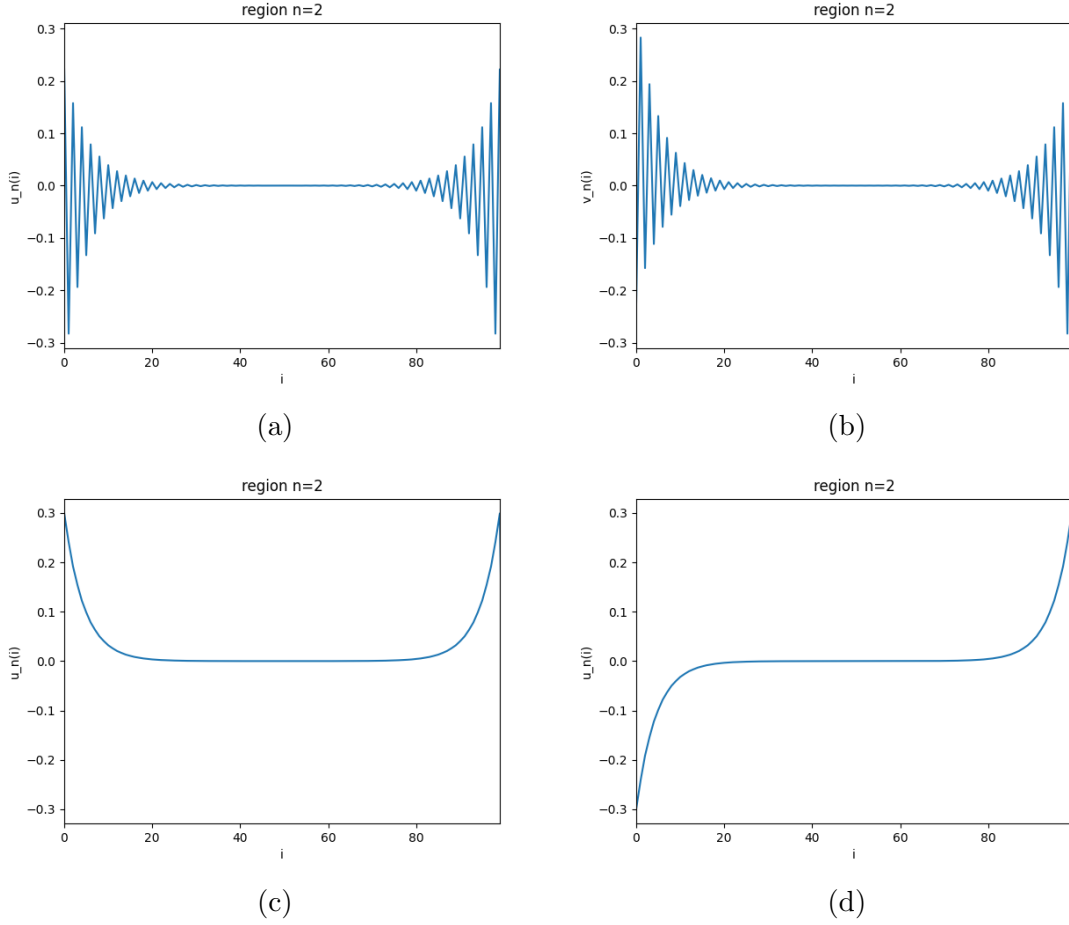


Figure 4: For $\lambda_1 = 0.05$ and $\lambda_2 = 1.5$, there are two Majorana zero modes. $u(i)$ and $v(i)$, which correspond to the first Majorana zero mode, are displayed in row number 1. The second Majorana zero mode, which is orthogonal to the first, is represented by the second row. Here we performed the numerical diagonalization With an open boundary condition for a lattice of $N = 100$ sites.

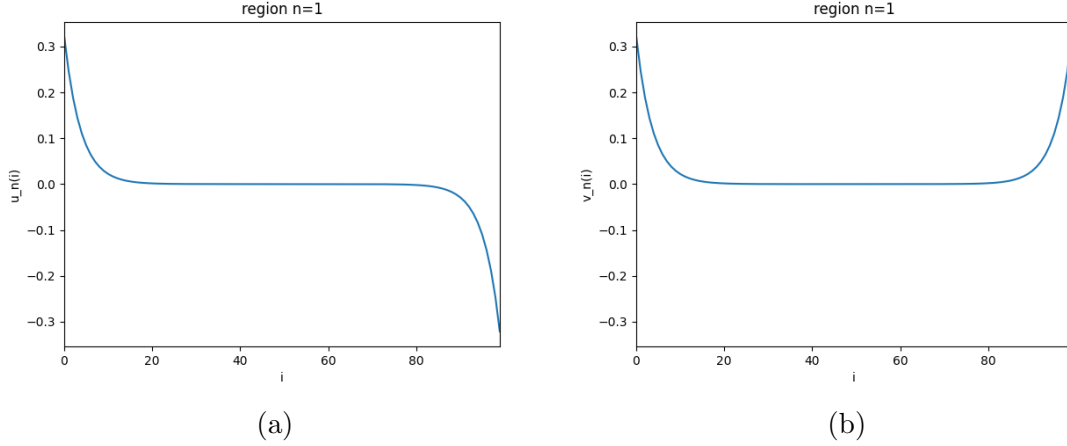


Figure 5: one Majorana zero mode for $\lambda_1 = 1.3$ and $\lambda_2 = 0.01$. This shows there is one MZM at the end of the chain where $u(i)$ is one and $v(i)$ is the one orthogonal to it. This is for $n = 1$ region in phase diagram Fig.2

What is a topological Invariant? A topological invariant is a quantity we can define that changes whenever there is a topological phase transition.

Winding number Winding number is a topological invariant quantity that changes value whenever there Topological phase transition. This is integer valued quantity hence we will know there is a phase transition whenever this quantity changes.

- Here we will define this winding number using the anderson pseudo spin vector that we can write as,

$$\vec{d}(k) = \Delta(k)\vec{y} + (\epsilon_k - \mu)\vec{z} \quad (22)$$

We have defined this vector only in the y-z plane here.

- When the time reversal symmetry is violated and the Anderson pseudo spin vector contains three components when the relative phase between these is a complex number. However, we won't talk about that here.
- lets define a unit vector which will help us in calculating the winding number

$$\hat{d}(k) = \frac{\vec{d}(k)}{\text{mod}(\vec{d}(k))} = \cos \theta_k \hat{y} + \sin \theta_k \hat{z} \quad (23)$$

- Therefor the integer winding number is just integral of this quantity given by,

$$W = \oint \frac{d\theta_k}{2\pi} \quad (24)$$

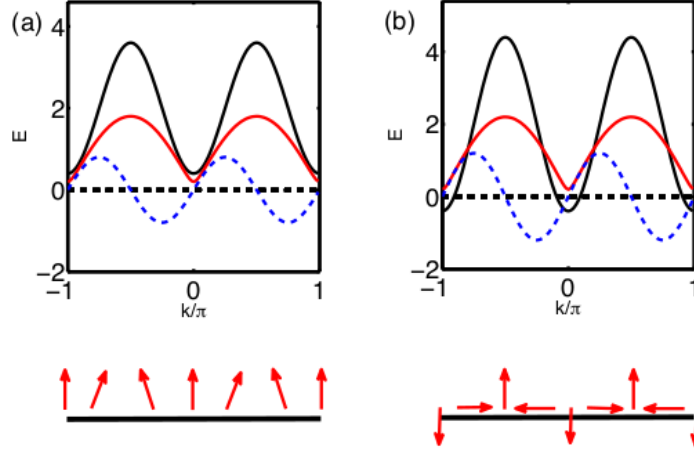


Figure 6: Topological phase transition in the vicinity of the point $\lambda_1 = 0$, $\lambda_2 = 1$. (a) $\lambda_2 = 1^-$, $\lambda_1 = 0$, (b) $\lambda_2 = 1^+$, $\lambda_1 = 0$. It is clear that in (a), the pseudospin does not wind along the 1d Brillouin zone, i.e. $W = 0$, whereas in (b) it winds twice, i.e. $W = 2$.

- This number indicates how many times the vector $\vec{d}(k)$ spins around the one-dimensional brillouin zone in the y-z plane.
- The winding number changes its value only when the energy gap closes i.e at a topological phase transition.
- The change in winding number also indicates the change in the number of normalizable majorana modes at the end of the chain after a transition.
- to study the transition at the critical point 'a' in the Fig.(2) across which the change in the number of normalizable Majorana modes at each end is 2. Here, the chain consists only of second-neighbor hopping and pairing. The winding number jumps from 0 to 2 across this transition. as shown in the Fig.(6)
- at the critical point 'e' in the Fig.(2) where $\lambda_1 = 1$ and $\lambda_2 = 0$, at this point the chain only consists of nearest neighbor hopping and pairing. and across this point change in winding number is equal to 1 which is identical to the change in the number of normalizable majorana fermion zero modes across this transition. as shown in the Fig.(7) given below.

8 What if time reversal symmetry is broken?

When we add a relative phase between the two order parameters in the BdG hamiltonian then we have,

$$\begin{pmatrix} \tilde{h} & \Delta \\ \tilde{\Delta}^\dagger & -\tilde{h} \end{pmatrix} \begin{pmatrix} \vec{u}_n \\ \vec{v}_n^* \end{pmatrix} = E_n \begin{pmatrix} \vec{u}_n \\ \vec{v}_n^* \end{pmatrix} \quad (25)$$

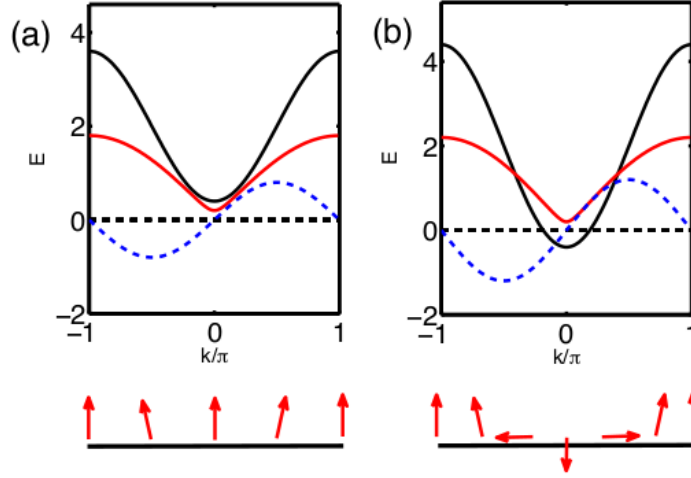


Figure 7: (a) $\lambda_1 = 1^-$ and $\lambda_2 = 0$; (b) $\lambda_1 = 1^+$ and $\lambda_2 = 0$, Evidently, the pseudospin vector in (a) does not wind along the 1d Brillouin zone hence we have $W = 0$, but in (b) it winds once hence $W = 1$.

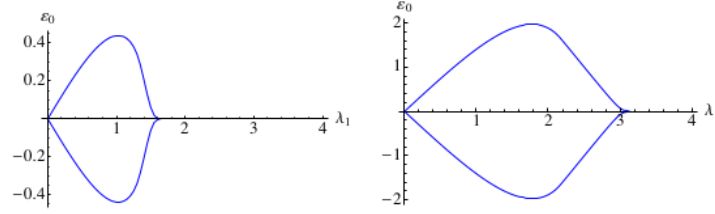


Figure 8: Splitting of $n = 2$ Majorana zero modes for the complex Hermitian BdG equation. Left: $\lambda_2 = 2.5$. Right: $\lambda_2 = -2.0$. The magnitude of ϵ_0 is the smallest eigenvalue

where the submatrices are,

$$\begin{aligned}\tilde{h}_{ij} &= \lambda_1 (\delta_{j,i+1} + \delta_{j,i-1}) + \lambda_2 (\delta_{j,i+2} + \delta_{j,i-2}) - 2\delta_{ij}, \\ \tilde{\Delta}_{ij} &= e^{i\theta} \lambda_2 (\delta_{i,j+2} - \delta_{i+2,j}) - \lambda_1 (\delta_{i-1,j} - \delta_{i+1,j})\end{aligned}\tag{26}$$

- The solution of this modified BdG Hamiltonian shows that the regions of the phase diagram which contain $n = 1$ Majorana zero mode remain robust while those containing $n = 2$ Majorana zero modes are in general destroyed, meaning that they are split.
- for an arbitrary value of θ the real and imaginary parts of Δ_{ij} will have contributions from both λ_1 and λ_2
- for $\theta = \pi/2$ the λ_2 term i.e the next nearest neighbor term will become purely imaginary and the λ_1 term i.e nearest neighbor term will be untouched.

9 Conclusion

So in this report we have studied the exactly solvable three spin extension TFIM , analytically solved it using J-W and BdG transformations , studied its phase diagram i.e understood the various regions inside it and also the critical lines and critical points where the system undergoes a phase transition. Next we have numerically simulated the majorana zero energy edge states that are present in this model near the critical lines and analyzed where there are one,two or zero MZMs. Next we focused on a topological invariant quantity winding number and found out that it changes its value whenever there is a topological phase transition and the value it changes by exactly equals the number MZMs in that region and then studied the longer ranged interactions of the MZMs localised at each end of the chain. Making it more physically plausible model leading to a more experimentally verifiable model.

10 Appendix A

The jordan wigner transformation is a technique using which we can transform our hamiltonian which is in terms pauli spin operators to fermionic operators.

The hamiltonian we wrote down previously given in Eq(??) is,

$$H = - \sum_i (g\sigma_i^x + \lambda_2\sigma_i^x\sigma_{i-1}^z\sigma_{i+1}^z + \lambda_1\sigma_i^z\sigma_{i-1}^x) \quad (27)$$

Now let us use the J-W equations to transform this hamiltonian.

The J-W equations which we will use are given by,

$$\begin{aligned} \sigma_i^x &= 1 - 2c_i^\dagger c_i, \\ \sigma_i^z &= \prod_{j<i} (1 - 2c_j^\dagger c_j) (c_i + c_i^\dagger), \end{aligned} \quad (28)$$

In terms of periodic boundary conditions the hamiltonian can be written as,

$$H = - \sum_{i=1}^N g\sigma_i^x + \sum_{i=1}^{N-1} (\lambda_2\sigma_i^x\sigma_{i-1}^z\sigma_{i+1}^z + \lambda_1\sigma_i^z\sigma_{i-1}^x) + \lambda_2\sigma_N^x\sigma_{N-1}^z\sigma_{N+1}^z + \lambda_1\sigma_N^z\sigma_{N-1}^x \quad (29)$$

Now let solve this term by term,

- First terms is,

$$g\sigma^x = g(1 - 2c_i^\dagger c_i) \quad (30)$$

- Second term is,

$$\begin{aligned} \lambda_2\sigma_j^x\sigma_{j-1}^z\sigma_{j+1}^z &= \lambda_2(1 - 2c_j^\dagger c_j) \left(- \prod_{l<j-1} (1 - 2c_l^\dagger c_l)(c_{j-1} + c_{j-1}^\dagger) \right) \left(- \prod_{k<j+1} (1 - 2c_k^\dagger c_k)(c_{j+1} + c_{j+1}^\dagger) \right), \\ &= \lambda_2 \left(- \prod_{l<j-1} (1 - 2c_l^\dagger c_l) \right)^{2\rightarrow \mathbf{I}} \left(1 - 2c_j^\dagger c_j \right)^{2\rightarrow \mathbf{I}} (c_{j-1} + c_{j-1}^\dagger)(c_{j+1} + c_{j+1}^\dagger), \\ &= \lambda_2 (c_{j-1}c_{j+1} + c_{j-1}c_{j+1}^\dagger + c_{j-1}^\dagger c_{j+1} + c_{j-1}^\dagger c_{j+1}^\dagger) \\ &= \lambda_2 (c_{j-1}^\dagger c_{j+1} + c_{j+1}c_{j-1} + h.c) \end{aligned} \quad (31)$$

- Third term is ,

$$\begin{aligned}
 \lambda_1 \sigma_i^z \sigma_{i-1}^z &= \lambda_1 \left(- \prod_{j < i} (1 - 2c_j^\dagger c_j) (c_i + c_i^\dagger) \right) \left(- \prod_{k < i-1} (1 - 2c_k^\dagger c_k) (c_{i-1} + c_{i-1}^\dagger) \right), \\
 &= \lambda_1 \left(\prod_{j < i-1} (1 - 2c_j^\dagger c_j) \right) \xrightarrow{2, \mathbf{I}} (1 - 2c_i^\dagger c_i) (c_i + c_i^\dagger) (c_{i-1} + c_{i-1}^\dagger), \\
 &= \lambda_1 \left(c_i + c_i^\dagger - \cancel{2c_i^\dagger c_i c_i} \xrightarrow{0} - 2c_i^\dagger c_i c_i^\dagger \right) (c_{i-1} + c_{i-1}^\dagger), \\
 &= \lambda_1 \left(c_i c_{i-1} + c_i c_{i-1}^\dagger + c_i^\dagger c_{i-1} + c_i^\dagger c_{i-1}^\dagger - 2c_i^\dagger c_{i-1} - 2c_i^\dagger c_{i-1}^\dagger \right), \\
 &= \lambda_1 \left(c_j^\dagger c_{j-1} + c_j^\dagger c_{j-1}^\dagger + h.c \right)
 \end{aligned} \tag{32}$$

Above overall we have used fermionic algebra where the anticommutation relation mainly is important.

- Now the Overall hamiltonian in terms of fermionic operators can be written down using above calculations, and the boundary terms we are ignoring.

$$H = -g \sum_{i=1}^N (1 - 2c_i^\dagger c_i) - \lambda_1 \sum_{i=1}^{N-1} (c_i^\dagger c_{i+1} + c_i^\dagger c_{i+1}^\dagger + h.c) - \lambda_2 \sum_{i=2}^{N-1} (c_{i-1}^\dagger c_{i+1} + c_{i+1} c_{i-1} + h.c) \tag{33}$$

11 Appendix B

The hamiltonian after J-W in Eq.(33) is given as,

$$H = -g \sum_{i=1}^N (1 - 2c_i^\dagger c_i) - \lambda_1 \sum_{i=1}^{N-1} (c_i^\dagger c_{i+1} + c_i^\dagger c_{i+1}^\dagger + h.c) - \lambda_2 \sum_{i=2}^{N-1} (c_{i-1}^\dagger c_{i+1} + c_{i+1} c_{i-1} + h.c) \quad (34)$$

Now applying the discrete fourier transform we can transform this hamiltonian to k-space using,

$$\begin{aligned} c_j &= 1/\sqrt{N} \sum_k e^{\frac{i2\pi kj}{N}} c_k, \\ c_j^\dagger &= 1/\sqrt{N} \sum_k e^{\frac{-i2\pi kj}{N}} c_k^\dagger \end{aligned} \quad (35)$$

Now lets apply the DFT term by term,

- Lets first consider the first subterm of the third term in Eq(33) i.e the λ_2 term

$$\begin{aligned} \sum_j c_{j-1}^\dagger c_{j+1} &= \frac{1}{N} \sum_j \left(\sum_l e^{-i\frac{2\pi l(j-1)}{N}} c_l^\dagger \right) \left(\sum_k e^{-i\frac{2\pi k(j+1)}{N}} c_k \right), \\ &= \frac{1}{N} \sum_l \sum_k \sum_j c_l^\dagger e^{i\frac{2\pi(k-l)j}{N}} e^{-i\frac{2\pi l}{N}} e^{-i\frac{2\pi k}{N}} c_k, \\ &= \frac{1}{N} \sum_l \sum_k c_l^\dagger N \delta_{lk} e^{-i\frac{2\pi(k+l)}{N}}, \\ &= \sum_k e^{i\frac{4\pi k}{N}} c_k^\dagger c_k \end{aligned} \quad (36)$$

- Now, the second subterm of the third term in Eq(33), i.e., the λ_2 term:

$$\begin{aligned} \sum_j c_{j+1} c_{j-1} &= \frac{1}{N} \sum_j \left(\sum_l e^{-i\frac{2\pi l(j+1)}{N}} c_l \right) \left(\sum_k e^{-i\frac{2\pi k(j-1)}{N}} c_k \right), \\ &= \frac{1}{N} \sum_l \sum_k \sum_j c_l e^{i\frac{2\pi(k+l)j}{N}} e^{i\frac{2\pi(l-k)}{N}} c_k, \\ &= \sum_k e^{-i\frac{4\pi k}{N}} c_{-k}^\dagger c_k \end{aligned} \quad (37)$$

The rest of the terms of λ_2 term will just be hermitian conjugates of above two terms.

- Let's consider the first subterm of the second term in Eq(33), i.e., the λ_1 term:

$$\begin{aligned}
\sum_j c_j^\dagger c_{j-1} &= \frac{1}{N} \sum_j \left(\sum_l e^{-i\frac{2\pi l j}{N}} c_l^\dagger \right) \left(\sum_k e^{i\frac{2\pi k(j-1)}{N}} c_k \right), \\
&= \frac{1}{N} \sum_l \sum_k \sum_j c_l^\dagger e^{i\frac{2\pi(k-l)j}{N}} e^{-i\frac{2\pi k}{N}} c_k, \\
&= \sum_k e^{-i\frac{2\pi k}{N}} c_k^\dagger c_k
\end{aligned} \tag{38}$$

- Let's consider the second subterm of the second term in Eq(33), i.e., the λ_1 term:

$$\begin{aligned}
\sum_j c_j^\dagger c_{j-1}^\dagger &= \frac{1}{N} \sum_j \left(\sum_l e^{-i\frac{2\pi l j}{N}} c_l^\dagger \right) \left(\sum_k e^{-i\frac{2\pi k(j-1)}{N}} c_k^\dagger \right), \\
&= \frac{1}{N} \sum_l \sum_k \sum_j c_l^\dagger e^{i\frac{2\pi(-k-l)j}{N}} e^{-i\frac{2\pi k}{N}} c_k^\dagger, \\
&= \sum_k e^{i\frac{2\pi k}{N}} c_{-k}^\dagger c_k^\dagger
\end{aligned} \tag{39}$$

The rest of the terms of λ_1 term will just be hermitian conjugates of above two terms.

Now the final hamiltonian using above calculations we can write as,

$$H(k) = \sum_k (2 - 2\lambda_1 \cos k - 2\lambda_2 \cos 2k) c_k^\dagger c_k + \sum_k \left(i\lambda_1 \sin k c_k^\dagger c_{-k}^\dagger + i\lambda_2 \sin 2k c_k^\dagger c_{-k}^\dagger + \text{h.c.} \right) \tag{40}$$

This is the hamiltonian after DFT and J-W transformation in k-space.

12 Appendix C

The Hamiltonian represented in k-space in Eq.(40) is not diagonalized. Now we would like to diagonalize it by applying another transformation, namely Bogoliubov-de-gennes transformation.

We define a new set of fermionic operators using the existing c_i 's

They follow the same fermionic algebra as before

$$\begin{aligned}\chi_k &= u_k c_k - v_k c_{-k}^\dagger, \\ \chi_{-k} &= u_k c_k - v_k c_k^\dagger\end{aligned}\tag{41}$$

By substituting above equations in Eq(40) and setting the off diagonal terms to zero we get the final diagonalized form and the excitation spectra is given by,

$$\epsilon_k = \pm \sqrt{1 + \lambda_1^2 + \lambda_2^2 + 2\lambda_1(1 - \lambda_2) \cos k - 2\lambda_2 \cos(2k)}\tag{42}$$